Loan Default Prediction

Comparing Classic ML and Neural Networks under Class Imbalance

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Final Project Group 15 · Home Credit Default Risk (Kaggle)

Business Problem and Motivation

Goal

Predict the probability that a loan applicant will default before making a lending decision.

Why it matters

- False negatives: approve borrowers who default
- False positives: reject creditworthy borrowers
- Models help balance risk vs. profitability

Key challenge
Default events are rare (~8%) $\Rightarrow$ strong class imbalance

From Imbalance to Research Question

Why imbalance matters

- Models can achieve high ROC AUC but still predict no defaults
- Default threshold (0.5) often fails in imbalanced settings

Common solutions

- Reweighting the loss (class weights)
- Resampling the data (e.g., SMOTE)
- Adjusting the decision threshold

Research Question

- Do neural networks respond to class imbalance handling techniques the same way as classic shallow learners?

Models compared

- Logistic Regression, Random Forest, HistGradientBoosting
- Shallow and Deep MLP

Dataset Overview

What the dataset represents

- Loan applications from Home Credit
- Focus on borrowers with limited credit history
- Tabular structure (Each row = one applicant)
- Goal: predict default risk

Main feature groups

- Demographics (age, family status)
- Employment and income
- Loan details (amount, annuity)
- External credit scores: `EXT_SOURCE_*`

Data Structure and Class Imbalance

Dataset characteristics

- Train data: **307,511**
- Test data: **48,744**
- Features: **122**

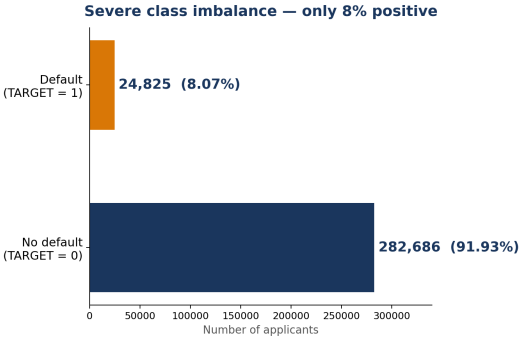
Target variable

- 1: default
- 0: repaid

Class imbalance

Defaults: only **8.07%** default

$\Rightarrow$ severe class imbalance



Classic ML Strategy

Why start with classic ML?

- The data is tabular: rows are applicants, columns are features
- Classical models are strong baselines for tabular prediction
- They provide a benchmark for the neural networks

Models tested

- **Logistic Regression**: strong baseline for probability prediction
- **Random Forest**: captures nonlinear patterns and reduces overfitting
- **Histogram Gradient Boosting**: captures complex interactions efficiently

Handling class imbalance

- Logistic Regression and Random Forest give more weight to default cases (`class_weight='balanced'`).
- Histogram Gradient Boosting used without class weights (boosting still captures rare cases well).

Classic ML Results

Model	Best hyperparameters	Val ROC AUC
HistGradientBoosting	`learning_rate`=0.05; `max_iter`=200; `leaf`=100	**0.7595**
Logistic Regression	`C`=0.1; `tol`=1e-5; `class_weight=balanced`	0.7487
Random Forest	`n_estimators`=200; `min_leaf`=1; `balanced`	0.7470

Key takeaways

- Histogram Gradient Boosting performs best (0.7595), suggesting boosted trees capture nonlinear interactions in tabular data effectively.
- Logistic Regression and Random Forest perform similarly, indicating strong linear signal alongside moderate nonlinearities.
- Random Forest shows signs of overfitting (train 0.96 vs val 0.747), while boosting remains more stable.

Neural Network Models

Architectures tested

Architecture	Hidden	Parameters	Val ROC AUC
Shallow MLP	$(64,)$	$\sim$15.7K	0.7405
Deep MLP	$(128, 64, 32)$	$\sim$42K	**0.7416**

Common setup

- ReLU activation, Adam optimizer, early stopping
- Tuned over `alpha` $\in \{10^{-4}, 10^{-3}\}$, `lr_init` $\in \{0.001, 0.005\}$
- Best for both: $\alpha = 10^{-3}$, `lr` $= 0.005$

Key takeaways

- Depth doesn't help: 0.7405 vs 0.7416 (within noise)
- Tripling parameters bought nothing
- Both NNs land below the classic learners

Imbalance Handling on the Best MLP

Three strategies compared (sklearn `MLPClassifier` does not support `class_weight`)

Method	Threshold	ROC AUC	PR AUC	Precision	Recall	F1
Baseline	0.500	0.742	0.217	0.00	0.00	0.00
SMOTE	0.500	**0.648**	0.134	0.16	0.21	0.18
Threshold tuning ⋆	0.139	0.742	0.217	0.22	**0.41**	**0.29**

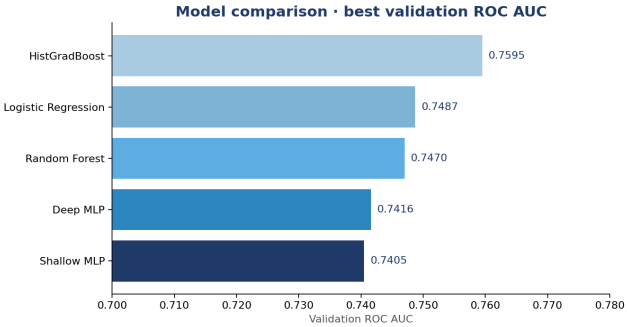
Key takeaways

- Baseline at 0.5: AUC 0.742 but recall **0%**
- SMOTE hurts: AUC drops 0.09
- Threshold tuning at 0.139 lifts recall to **41%**

Headline

Choosing the threshold matters more than rebalancing the data.

Combined Leaderboard



Final ranking

Model	Val ROC AUC
HistGradBoost	**0.7595**
Logistic Reg.	0.7487
Random Forest	0.7470
Deep MLP	0.7416
Shallow MLP	0.7405

Tree boosting wins by ∼1 AUC point — consistent with the literature on tabular data.

Why NN Underperforms on Tabular Data

Both MLPs land below LR / RF / HGBC. Why?

1. **Tabular favors trees, not NN**
   - NN excels on image, text, audio — continuous, structured inputs
   - Trees split on individual features cleanly; NN must weight all at once
2. **Feature signal is already linear-friendly**
   - `EXT_SOURCE_*` are pre-engineered credit scores — LR / trees use them directly
   - NN must rediscover the linear relationship through nonlinear layers
3. **One-hot encoding penalizes NN**
   - 16 categorical columns → 140 sparse 0/1 columns (70% of features)
   - Trees handle sparse binary inputs natively; MLP must learn weights on all
4. **sklearn MLP has no `class_weight`**
   - LR / RF apply `class_weight='balanced'` in the loss; MLP cannot

Consensus in the literature

Tree-based models still outperform deep learning on tabular data (Grinsztajn et al., NeurIPS 2022).

Conclusions

Best model overall

- HGBC wins (val AUC = **0.7595**) — nonlinear tree boosting captures feature interactions
- LR and RF close behind — LR benefits from strong linear predictors (`EXT_SOURCE_*`)
- Deep MLP (0.7416) $\approx$ Shallow MLP (0.7405) — depth doesn't help
- Both NNs lag classic shallow learners by $\sim$1–2 AUC points

Imbalance handling on the MLP

- Threshold tuning lifts recall **0% $\rightarrow$ 41%**; SMOTE hurts AUC
- Choosing the threshold matters more than rebalancing the data

Limitations

- Only main `application_*.csv` tables (auxiliary tables would add several AUC points)
- Single train/val split, not full k-fold CV

Recording

<<INSERT RECORDING LINK BEFORE SUBMISSION>>

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